



UMD DATA605 - Big Data Systems

Storage DB internals

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v1.1

UMD DATA605 - Big Data Systems

Outline

- Storage
 - Physical storage
 - Storage hierarchy
 - Magnetic disks / SSDs
 - RAID
 - DB internals

- Sources: Silberschatz et al. 2020

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 - **Storage hierarchy**
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- Sources: Silberschatz et al. 2020, Chap 12, Physical Storage Systems

Storage Characteristics

- Storage media presents a trade-off between:
 - speed of access (e.g., 500-3,500MB / sec)
 - cost per unit of data (e.g., 50 USD / TB)
 - medium reliability
- Volatile vs non-volatile storage
 - **Volatile**: loses contents when power switched off
 - **Non-volatile**: can survive failures and system crashes
- Sequential vs random access
 - **Sequential**: read the data contiguously
`SELECT * FROM employee`
 - **Random**: read the data from anywhere at any time
`SELECT * FROM employee`
`WHERE name LIKE '\\\_a\\\_b'`
- Need to know how data is stored in order to optimize access



Storage Hierarchy

Various storage can be organized in a hierarchy according to (decreasing) speed and cost

- **Cache**

- Fastest and most costly
- ~MBs on chip
- DB developers do pay attention to cache effects

- **Main memory**

- Up to 100s of GBs
- Typically can't store the entire DB
- Volatile

- **Flash memory / SSDs**

- More expensive than RAM but less than magnetic disk
- Non-volatile, random access

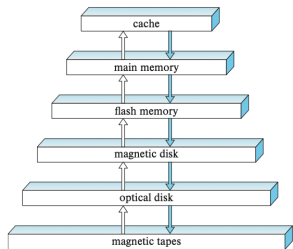
- **Magnetic disk**

- Long-term on-line storage
- Non-volatile (can survive failures and system crashes)

- **Optical disk (CD, Blue Ray)**

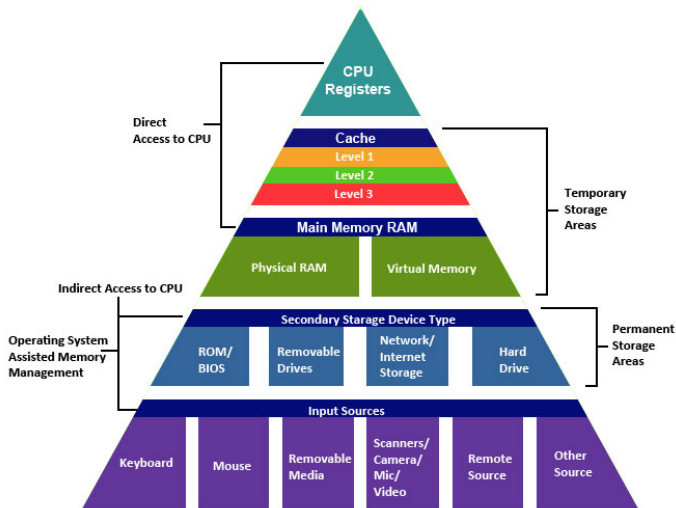
- Mainly read-only

- **Magnetic tapes**



- **Primary storage:** cache, main memory
- **Secondary (or online):** flash memory, magnetic disk
- **Offline:** optical, magnetic tape

Storage Hierarchy



source: <http://cse1.net/recaps/4-memory.html>

How Important is Memory Hierarchy?

- Trade-offs shifted drastically over last 10-15 years
- **Innovations:**
 - Fast network, SSDs, and large memories
 - However, the volume of data is growing rapidly
- **Observations:**
 - Faster to access another computer's memory through network than accessing your own disk
 - Cache is playing more and more important role
 - In-memory DBs
 - Enough memory that data often fits in memory of a cluster of machines
 - Disk considerations less important
 - Still disks are where most of the data lives today
- Similar reasoning for algorithms
 - Best algorithm depends on the technology available

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Connecting disks to a server

- **Disks** (magnetic and SSDs) can be **connected to computer**:
 - Through high-speed bus interconnections; or
 - Through high-speed network
- **Through a high-speed interconnection**
 - Serial ATA (SATA)
 - Serial Attached SCSI (SAS)
 - NVMe (Non-volatile Memory Express)
- **Through high-speed networks**
 - Storage Area Network (SAN): iSCSI, Fiber Channel, InfiniBand
 - **Network Attached Storage (NAS)**
 - Provides a file-system interface (e.g., NFS)
 - Cloud storage: Data is stored in the cloud and accessed via an API, Object store, High latency

Magnetic Disks

- 1956
 - IBM RAMAC
 - 24" platters
 - 5 million characters



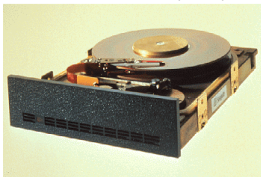
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Magnetic Disks

- 1979
- SEAGATE
- 5MB

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- 1998
- SEAGATE
- 47GB

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- 2006
- Western Digital
- 500GB



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Magnetic Disks: Components

- **Platters**

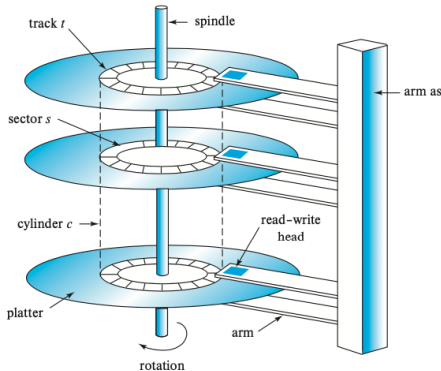
- Made of rigid metal covered with magnetic material on both surfaces
- It spins at 5400 or 9600 RPM
- *Tracks* subdivided into *sectors* (smallest unit read or written, with a checksum)

- **Read-write heads**

- Store information magnetically on the disk
- Spinning creates a cushion that maintain the heads a few microns from the disk surface
- *Cylinder* is the i -th tracks of all the platters (can be read / written together)

- **Arm**

- Move all the heads along the disks



- **Disk controller**

- Accepts high level commands to

Magnetic Disks: Current Specs



- **Capacity**
 - 10 terabyte and more
- **Access time**
 - = Time to start reading data
 - Seek time
 - = Move the arm across cylinders (2-20ms)
 - Rotational latency time
 - = Wait for sector to be accessed (4-12ms)
- **Data-transfer rate**
 - Once the data is reached the transfer begins
 - Transfer rate = 50-200MB / secs
 - Sector (disk block) = logical unit of storage (4-16KB)
 - Sequential access = when the blocks are on the same or adjacent tracks
 - Random access = each request requires a seek
 - IOPS = number of random single block accesses in a second (50-200 IOPS)
- **Reliability**
 - Mean time to failure (MTTF) = the amount of time that on average the system can run continuously without a failure
 - Lifespan of an HDD is ~5 years

Accessing Data Speed

- **Random data transfer rates**

- = how long it takes to read a random sector
- It has 3 components
 - Seek time: Time to seek to the track (Average 4 to 10ms)
 - Rotational latency: Waiting for the sector to get under the head (Average 4 to 11ms)
 - Transfer time: Time to transfer the data (Very low)
- About 10ms per access
 - So if randomly accessed blocks, can only do 100 block transfers (100 / sec x 4 KB per block = 400 KB/s)

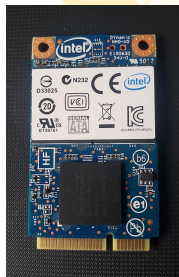
- **Serial data transfer rates**

- = rate at which data can be transferred (without any seek)
- 30-50MB/s to up to 200MB/s

- **Seeks are bad!**

Solid State Disk (SSD)

- Mainstream around 2000s
 - Like non-volatile RAM (NAND and NOR)
 - **Capacity**
 - 250, 500 GBs (vs 1-10 TB for HDD)
 - **Access time**
 - Latency for random access is 1,000x smaller than HDD
 - E.g., 20-100 us (vs 10 ms HDDs)
 - Multiple random requests (e.g., 32) in parallel
 - 10,000 IOPS (vs 50/200 for HDDs)
 - Require to read an entire “page” of data (typically 4KB)
 - Equivalent to a block in magnetic disks
 - **Data-transfer rate**
 - 1 GB/s (vs 200 MB/s HDD)
 - Typically limited by the interface speed
 - Reads and writes are ~500MB/s for SATA and 2-3 GB/s for NVMe
 - Lower power consumption than HDDs
 - Writing to SSD is slower than reading (~2-3x)
 - It requires erasing all pages in the block
 - **Reliability**
 - There is a limit to how many times a flash page can be erased (~1M times)
- Better than an HDD from any point of view, but more expensive per GB



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- Storage
 - Physical storage
 - Storage hierarchy
 - Magnetic disks / SSD
 - **RAID**
- DB internals

RAID

- RAID = Redundant Array of Independent Disks
- **Problem**
 - Storage capacity has been growing exponentially
 - Data-storage requirement (e.g., web, DBs, multimedia applications) has been growing even faster
 - You need a lot of disks
 - MTTF between failure of any disk get smaller (e.g., days)
 - If we store a single copy of the data, the frequency of data loss is unacceptable



- **Observations**

- Disks are very cheap
- Failures are very costly
- Use “extra” disks to ensure reliability
 - Store data redundantly
 - If one disk goes down, the data still survives
- Bonus: allow faster access to data

- **Goal**

- Expose a logical view of a single large and reliable disk from many unreliable disks
- Different RAID “levels” (reliability vs performance)

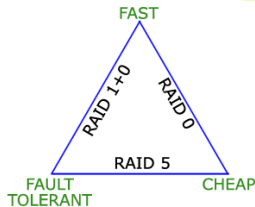
Improve Reliability / Performance with RAID

- **Reliability**

- Use redundancy
 - Store the same data multiple times: E.g., mirroring (aka shadowing)
 - If a disk fails, the data is not lost but it can be reconstructed
 - Increased MTTF
- Assumption: independence of disk failure
 - Power failures and natural disasters
 - As disks age, probability of failure increases together

- **Performance**

- Parallel access to multiple disks: E.g., mirroring, Increase number of read requests
- Striping data across multiple disks: Increase transfer rate



RAID Levels

- **RAID 0: No redundancy**

- Array of independent disks
- Same access-time
- Increased transfer rate

- **RAID 1: Mirroring**

- Make a copy of the disks
- If one disk fails, we have a copy
- Reads: can go to either disk, so higher data rate possible
- Writes: need to write to both disks

- **RAID 2: Memory-style error correction**

- Use extra bits so we can reconstruct
- Superseded by RAID 5

- **RAID 3: Interleaved parity**

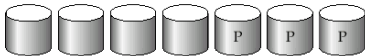
- One disk contains "parity" for the main data disks
- Can handle a single disk failure
- Little overhead (only 25% in the above case)



(a) RAID 0: nonredundant striping



(b) RAID 1: mirrored disks



(c) RAID 2: memory-style error-correcting codes



(d) RAID 3: bit-interleaved parity



(f) RAID 5: block-interleaved distributed parity

Choosing a RAID Level

- Main choice between RAID 1 and RAID 5
- **RAID 1 better write performance**
 - E.g., to write a single block
 - RAID 1: only requires 2 block writes
 - RAID 5: 2 block reads and 2 block writes
 - Preferred for applications with *high update rate* and *small data* (e.g., log disks)
- **RAID 5 lower storage cost**
 - RAID 1: 2x more disks
 - RAID 5 is preferred for applications with *low update rate* and *large amounts of data*



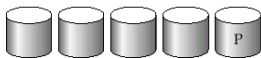
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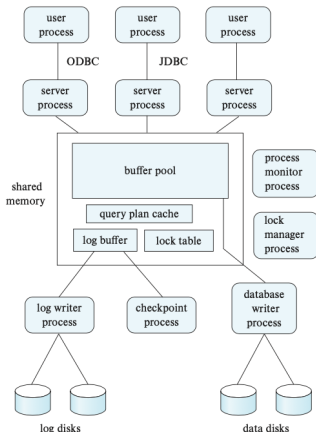
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- Sources: Silberschatz et al. 2020, Chap 13: Data Storage Structures

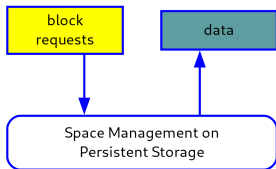
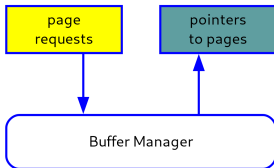
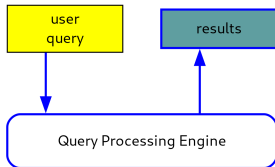
(Centralized) DB Internals

- User processes
 - Issue commands to the DB
- Server processes
 - Receive commands and call into the DB code
- Process monitor process
 - Monitor DB processes
 - Recover processes from failures
- Lock manager process
 - Lock grant / release
 - Deadlock detection
- Database writer process
 - Output modified buffer blocks to disk on a continuous basis
- Log writer process
 - Output log records to stable storage
- Checkpoint process
 - Perform periodic checkpoints
- Shared memory

- Contain all shared data



DB Internals



- **Query Processing Engine**

- Given a user query, decide how to “execute” it
- Specify sequence of pages to be brought in memory
- Operate upon the tuples to produce results

- **Buffer Manager**

- Bring pages from disk to memory
- Manage the limited memory

- **Storage hierarchy**

- How are tables mapped to files?
- How are tuples mapped to disk blocks?